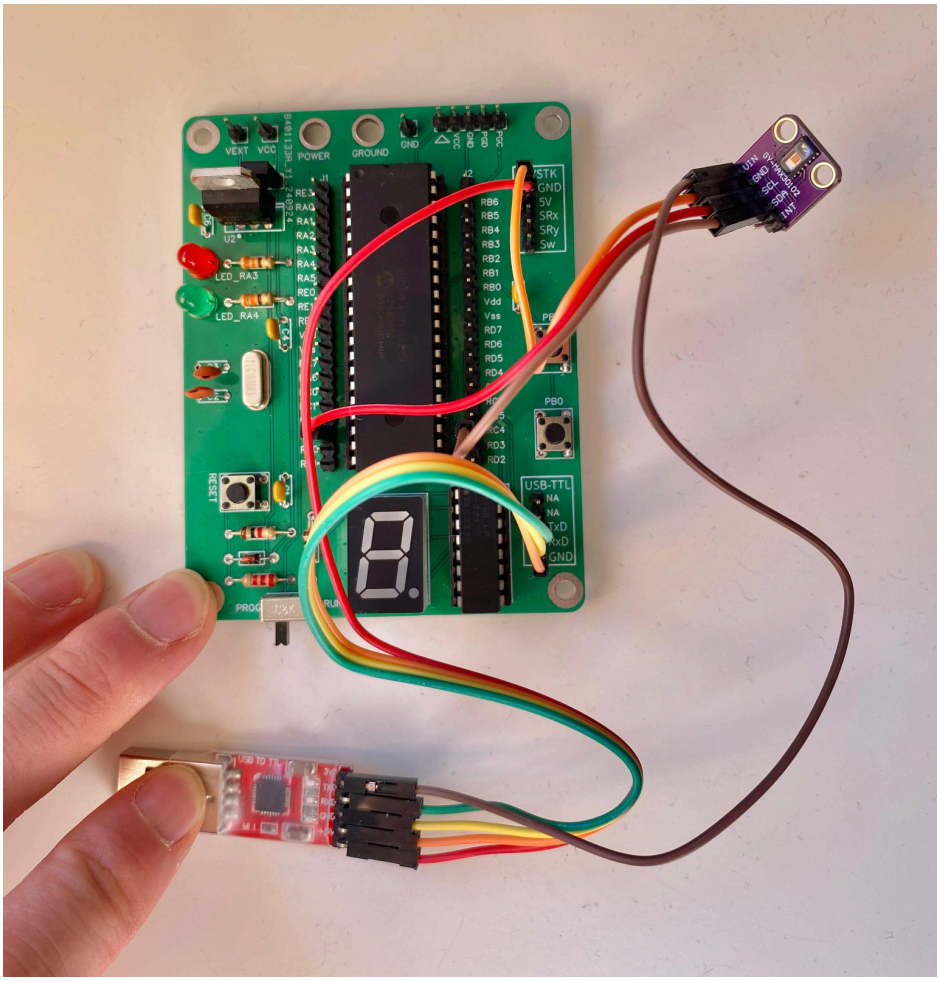
laSalle
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Pulse oximeter

Digital Electronic Systems



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Introduction and Objective

This project aims to develop a **pulse oximeter** system, with the primary goal of addressing the next needs. The motivation behind choosing this work was twofold. First, it allowed us to integrate theoretical and practical knowledge from the subject *Digital Electronic Systems*, specifically involving embedded systems, biomedical signal acquisition, processing and display, and communication protocols. Second, the topic offers real-world relevance with high applicability in modern healthcare. Unlike other possible microcontroller-based projects, this one provides direct insight into biomedical engineering and human-centered instruments.

The measurement of vital signs like heart rate and blood oxygen saturation plays an important role in medical diagnostics and personal health monitoring, both in hospitals and domestic environments. These physiological parameters are critical in the early detection of cardiovascular and respiratory conditions and in the management of diseases such as heart failure, COPD, COVID-19 and sleep apnea.

With the rise of wearable technology and remote patient tracking, the demand for compact and energy-efficient devices for capturing biological indicators has increased substantially.



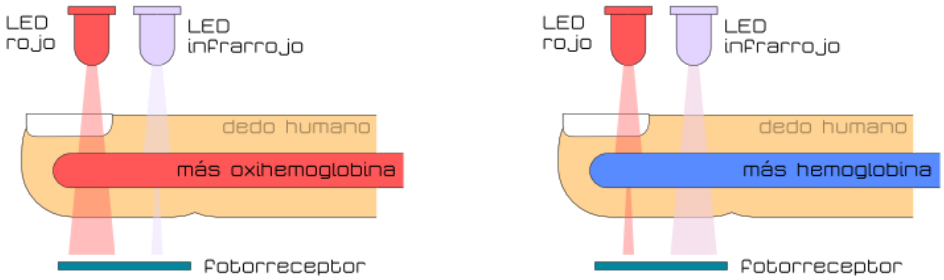
General Overview

The system is composed of three main components, each interconnected through specific communication peripherals. The **MAX30102** optical **sensor** is responsible for the acquisition of the data and communicates with the **PIC18F4321 microcontroller** via the **I²C** protocol. The microcontroller acts as the central unit, acquiring data, processing it and managing communication, so it transmits the information to the **computer** via **UART** interface through a **CP2102 USB-to-UART Converter**. Finally, the computer receives and displays the data in real time via **HyperTerminal**. Together, they establish the communication pathway of the device from sensor to microcontroller and finally to the user, providing clear and accessible readings of **heart rate** and **blood oxygen saturation** to be analyzed.

Theoretical Background

Principles of Pulse Oximetry

Pulse oximetry is a non-invasive technique that estimates the **heart rate** (HR) and **blood oxygen saturation** (SpO_2) by analyzing some characteristics of the pulsatile blood flow. The sensor emits **red light** (around 660 nm) and **infrared light** (around 880–940 nm) through a tissue, typically a fingertip or earlobe. The amount that is **absorbed** by the **blood** varies depending on the **oxygen** level in the **hemoglobin** protein present inside red blood cells. **Oxygenated hemoglobin** absorbs more **infrared light** and less **red light**, while **deoxygenated hemoglobin** does the opposite. By measuring the ratio of the two types of light over time and applying the fundamental physical principle named Beer-Lambert law, the device can estimate the SpO_2 level. Simultaneously, the changes in the detected intensity due to pulsatile blood flow are used to determine the heart rate.

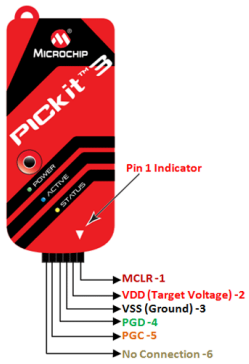


Characteristics of the Components: PIC18F4321 microcontroller, PicKit3, MAX30102 sensor and CP2102 USB-to-UART Converter

The **PIC18F4321** is an 8-bit **microcontroller** from Microchip, based on the PIC18 family, designed for high-performance embedded applications. It features a range of integrated peripherals, including I²C and UART communication modules, digital I/O, timers and analog-to-digital converters (ADC). Its code is programmed using **MPLAB X IDE** in **C language**, providing precise control over low-level functions. In this project, it is the main controller, responsible for managing data acquisition from the MAX30102 sensor via I²C protocol, processing and transmission to the computer via UART interface.

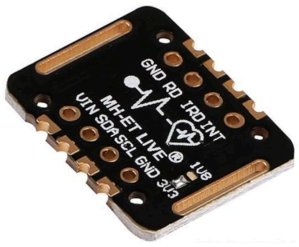


The **PICkit 3** is a programming tool used to upload the compiled code into the PIC18F4321 microcontroller. It is used during the development stage, not as part of the final operating system.



The **MAX30102** is a **sensor** designed for heart rate and SpO_2 measurement, with internal **analog-to-digital conversion**. It integrates two LEDs, that are the red and infrared lights, a photodetector, optical elements, and low-noise electronics. It also includes onboard algorithms for data acquisition and filtering. Both **lights** are **emitted**, the photodetector captures the **reflected light**, which is first amplified and filtered as an analog signal, then digitized internally by the ADC. Its advantages include small size, low power consumption and a high signal-to-noise ratio, which makes it well-suited for portable and battery-powered implementations.





The **CP2102 USB-to-UART Converter** enables communication by converting UART messages from the PIC18F4321 microcontroller into USB messages understood by the computer, allowing transmission to different tools.



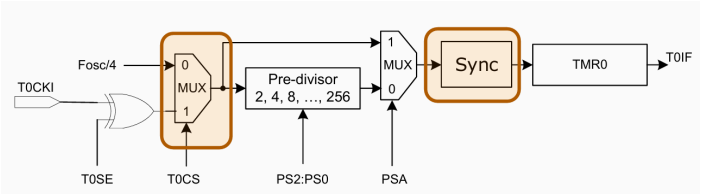
Basics of the Peripherals: Timers, I²C and UART communication and HyperTerminal use

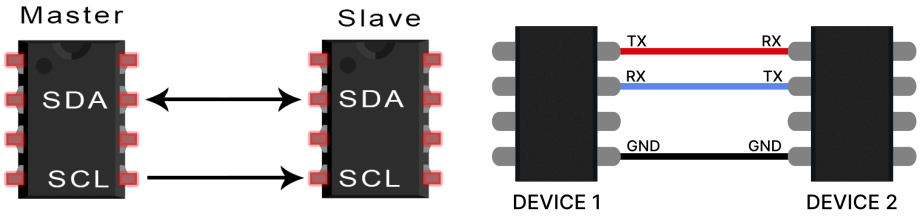
Timers are internal counters in microcontrollers used to measure time or generate precise delays. In this task, the PIC18F4321 microcontroller uses the Timer0. It can be configured in 8-bit or 16-bit mode, with selectable prescalers and clock sources. This setup allows it to overflow, triggering time-based events without requiring constant CPU attention and intervention.

I²C (Inter-Integrated Circuit) is a synchronous, multi-master, multi-slave, packet-switched protocol widely used for short-distance serial data communication between microcontrollers and peripherals. It uses only two wires: SCL (clock) and SDA (data), making it ideal for small-scale architectures. In our task, it enables the PIC18F4321 microcontroller to communicate efficiently with the MAX30102 sensor.

UART (Universal Asynchronous Receiver/Transmitter) is an asynchronous interface for serial data communication used point-to-point. It operates through two lines: **TX** (transmit) and **RX** (receive). It is used in this task to communicate the PIC18F4321 microcontroller to a computer.

HyperTerminal is a software application for serial communication. In the task, it allows the computer to receive and display the data.



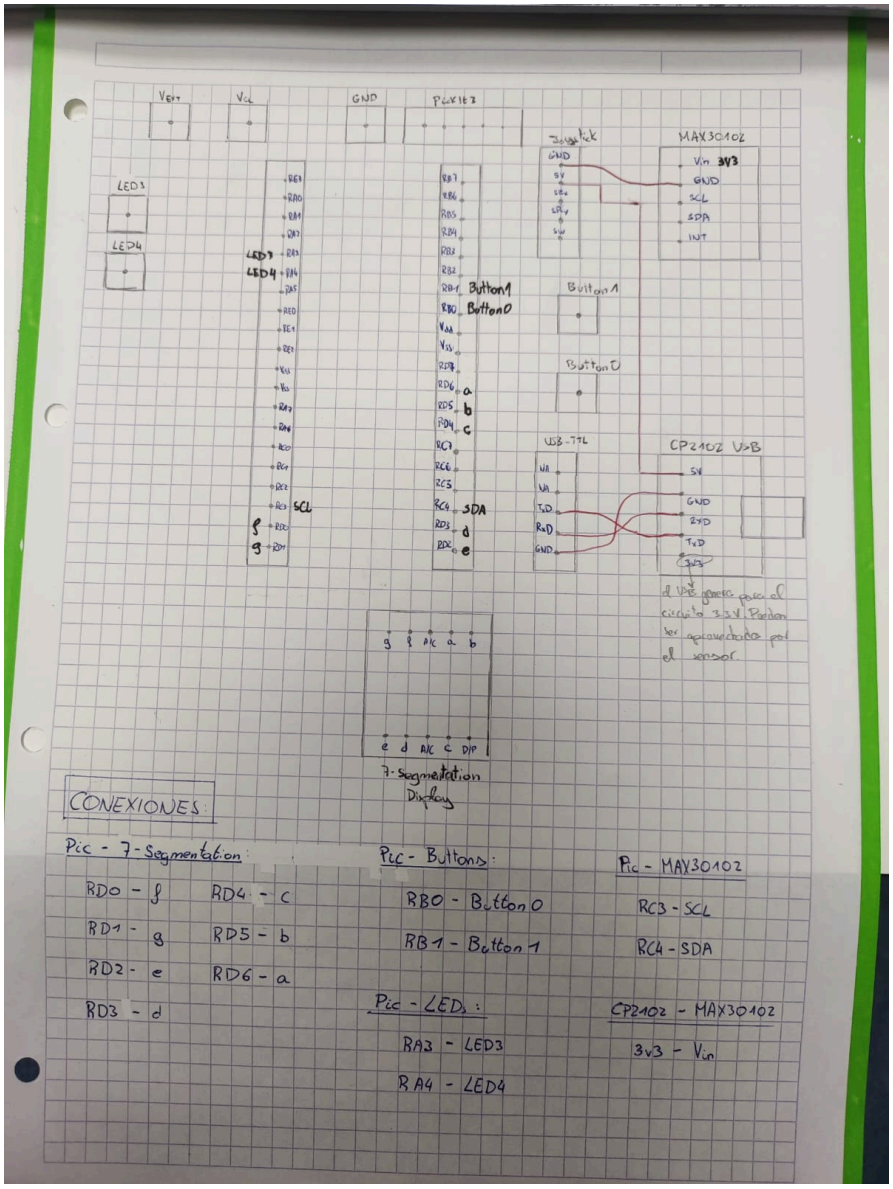




Hardware Design

Diagram, Wiring and Pinout

This is the initial diagram of our device that we did in order to be able to consult visually the wiring and pinout in a fast way:



The hardware design is built around a PIC18F4321 microcontroller, which connects with a PicKit3, a MAX30102 sensor, a CP2102 USB-to-UART Converter, two LEDs, push buttons and a 7-segment display. The connections are arranged to ensure proper communication of each module.

PIC18F4321 – PicKit 3

- PROG → MCLR.
- VCC → VDD.
- GND → VSS.
- PGD → PGD.
- PGC → PGC.

PIC18F4321 – MAX30102 (I²C protocol):

- RC3 (SCL) → SCL.
- RC4 (SDA) → SDA.

PIC18F4321 – CP2102 (UART interface):

- RC7 (SIORx)/USB-TTL: TxD → TxD.
- RC6 (SIOTx)/USB-TTL: RxD → RxD.

PIC18F4321 – LEDs:

- RA3 → LED3.
- RA4 → LED4.

The LEDs serve as status indicators.

PIC18F4321 – Push Buttons:

- RB0 → Button 0.
- RB1 → Button 1.

These push buttons are digital inputs used for user interaction.

PIC18F4321 – 7-Segment Display:

- RD0 → f.
- RD1 → g.
- RD2 → e.
- RD3 → d.
- RD4 → c.
- RD5 → b.
- RD6 → a.

The 7-segment display allows control for the visual output.

Power (Vcc) and Ground (0V)

A common power and ground connection is established, to maintain a consistent reference level across the system.

In the case of the MAX30102, it requires only 3.3V at its Vin pin, which is supplied from the CP2102 3V3 pin. This has provided regulation to ensure the voltage is safe for the sensor.

Software and Firmware Development

The software and firmware development of the device focuses on programming the PIC18F4321 microcontroller to acquire, process, and transmit data from the MAX30102 sensor to the computer. This is achieved using MPLAB X IDE through a modular code structure written in C language. The project relies on a main program and several custom files that manage different aspects of the system, such as I²C protocol, UART interface and HyperTerminal. Each one plays a specific role in ensuring efficient and organized functionality.

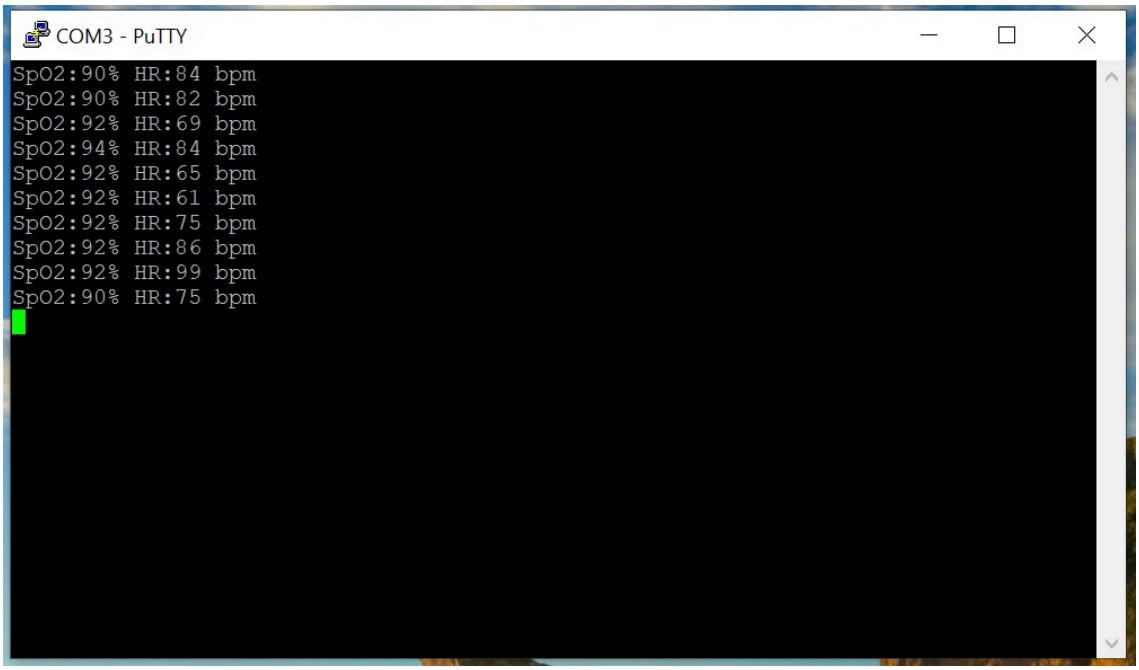
The **newmain.c** file contains the central logic of the system. It begins by configuring the microcontroller's ports and initializing all peripherals, including timers, UART communication and the I²C interface. The MAX30102 sensor is then initialized to prepare for data collection. Inside the main loop, the program waits for a push button press, that when detected and debounced, the system turns on an LED, reads red and infrared light values from the sensor through I²C, and transmits them to the PC via UART. After sending multiple samples, the LED turns off, providing simple user feedback. This structure enables responsive and organized control of biomedical signal capture and processing.

The **TTimer** module provides a software timer system using Timer0 of the PIC18F4321 microcontroller. It allows managing up to five independent timers by counting time intervals without blocking the CPU. It is configured to overflow every 10 ms, triggering an action. Functions are provided to initialize timers, update, retrieve and reset their tick counts, and handle interrupts efficiently. This setup enables precise timing control for multiple tasks like LED blinking or push button debouncing.

The **TI2C** module handles the I²C protocol between the PIC18F4321 microcontroller and compatible devices such as the MAX30102 sensor. It initializes the microcontroller as a master and includes functions for sending and receiving data through the bus. The code implements basic operations like start/stop conditions, byte write/read, and acknowledgment handling. It also contains blocks specifically for the sensor, as an initialization by writing to configuration registers and reading data from its FIFO memory. This setup allows efficient and structured acquisition of red and infrared light signals needed for pulse oximetry measurements.

The **TSIO** module defines the UART interface between the PIC18F4321 microcontroller and the computer. The implementation initializes the EUSART module with 19200 baud and high-speed mode using a 10 MHz clock in asynchronous mode. Key functions include sending data and receiving it through the TX and RX lines.

The **THyperT** module implements a simple way to format data from the PIC18F4321 microcontroller to the computer via HyperTerminal. It includes functions that process raw readings from the MAX30102 sensor, estimate heart rate (HR) and blood oxygen saturation (SpO_2), and transmit the results over UART in a human-readable format. Although the calculations are simplified, this module provides a clear and structured interface protocol for interpreting physiological signals and delivering real-time data.



Testing and Results

Methodology

To verify correct system operation, we tested each communication interface separately and then integrated the full data flow.

We implemented a timer to replace delay functions. This allowed precise control of tasks like LED blinking and push button debouncing, enhancing efficiency.

Once done, we checked I²C communication between the PIC18F4321 microcontroller and the MAX30102 sensor. We set it up in master mode. Using an interrupt-driven state machine, the microcontroller manages the data delivery phases smoothly, including start, addressing, reading, and stop conditions. We also confirmed the timing with an oscilloscope. This ensures reliable and continuous results.

We validated the UART interface communication between the PIC18F4321 microcontroller and the computer using the CP2102 USB-to-UART Converter. First, we periodically sent test characters ('U'). We confirmed the bit timing with an oscilloscope. Reception was tested by looping Tx and Rx, toggling LEDs and displaying received characters in the 7-segment display.

Finally, we integrated both protocols to acquire data from the sensor, process it and display it through HyperTerminal communication.

Outcomes

When one of the push buttons is pressed, which is synchronized with the timer to ensure accurate timing, and a LED turns on, the data acquisition from the MAX30102 sensor begins. Then, the PIC18F4321 microcontroller successfully read sensor data from the MAX30102 sensor using I²C communication. After that, UART was set up to send and receive data at 19200 bps. Finally, all parts were integrated so data is sent and displayed in real time through HyperTerminal, providing clear user feedback.

Comparison with Commercial devices

Due to time and resource limitations, the system was not directly compared with a certified commercial pulse oximeter device. However, typical heart rate and blood oxygen saturation values were observed and appeared reasonable under resting conditions (~70–100 bpm, ~90–98% SpO_2), suggesting functionality and basic reliability.

Discussion

Several challenges emerged during the development and testing phases. One of the main difficulties was the integration of the MAX30102 sensor, since implementing data extraction proved more complexity than expected.

Although the whole communication was established, we believe that due to incorrect initialization of both the I²C protocol and the UART interface, it affected the overall behavior of the sensor, which did not respond as expected during the initial tests.

Another source of inaccuracy and variability in the results was noise in the signal caused by finger movement and ambient light interference. In the current version, no digital filtering is implemented, which makes the data less stable.

For future improvements, the most immediate step would be to implement more accurate calculations for heart rate (HR) and blood oxygen saturation (SpO_2), using signal processing techniques such as peak detection or averages.

Additionally, adding an OLED or LCD screen would provide more detailed feedback without relying on a PC. Furthermore, a future work could be incorporating Bluetooth Low Energy (BLE), that would allow wireless transmission of data to mobile applications. Integrating signal processing algorithms directly in the microcontroller or offloading them to a smartphone could also enhance the system.

Conclusions

This project successfully designed, implemented, and tested the key hardware and software elements of a pulse oximeter device based on the PIC18F4321 microcontroller.

The development of a modular and structured firmware enabled effective management and streamlined the overall system operation. By integrating timers to replace simple delays, we achieved an improvement in timing. We established reliable communication protocols using I²C to acquire sensor data from the MAX30102 sensor and UART to transmit that data to a computer for real-time visualization via HyperTerminal.

The system presents strong potential as an educational and experimental platform for biomedical signal capture. It provides a solid foundation for further enhancements, including advanced signal processing, integration of visual displays and wireless communication technologies. Its applications span from academic laboratories to the early development of medical prototypes and diagnostic equipment.